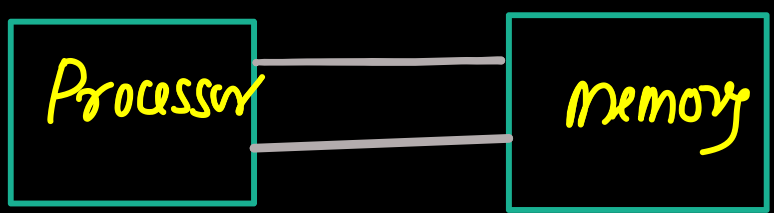
Memory systems. (Note)

— A computer based system consists of a processor and memory.

— memory store the information, which include data and instruction



— processor fetch the instruction from the memory, decode them and execute them. During the execution, the CPU also reads from memory and write the result back to the memory.

→ processor communicates with memory using a memory interface consists of buses and control signals.

→ Two fundamental operations performed on memory are read and write

Read operation

- processor places address on address buss
- processor assert read control signal

→ memory system return the data on the read data bus.

Write operation

- processor place the address on the address Bus
- processor place the data on the write data Bus
- processor assert control signal

→ memory store the data

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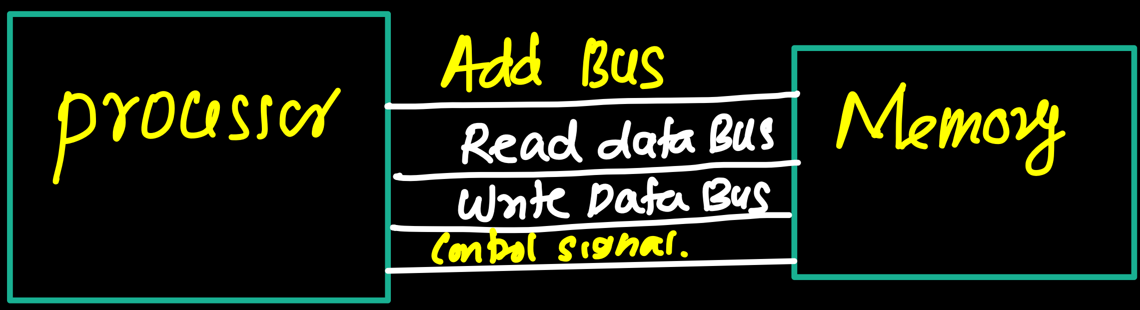


Speed Gap btw processor and Memory.

- processor is a very fast device
- processor speed is given by clock frequency.

eg: If $f_c = 2GHz = 2\times10^9$ Hz

$$T = \frac{1}{f_c} = \frac{1}{2\times10^9} = \underline{\underline{0.5ns}}$$

ie clock cycle time = 0.5ns.,

CPU can complete one task in 0.5ns.

→ Processor needs to access the memory, But memory access time is much larger value. ie memory is comparatively slow device.

→ CPU waste many cycles for memory.

- We need to bridge the gap processor and memory.

Design factor of memory

While designing the memory system, three important factors are considered

1. access time
2. Cost
3. Memory size

In general Faster memory is more expensive and smaller size but slower

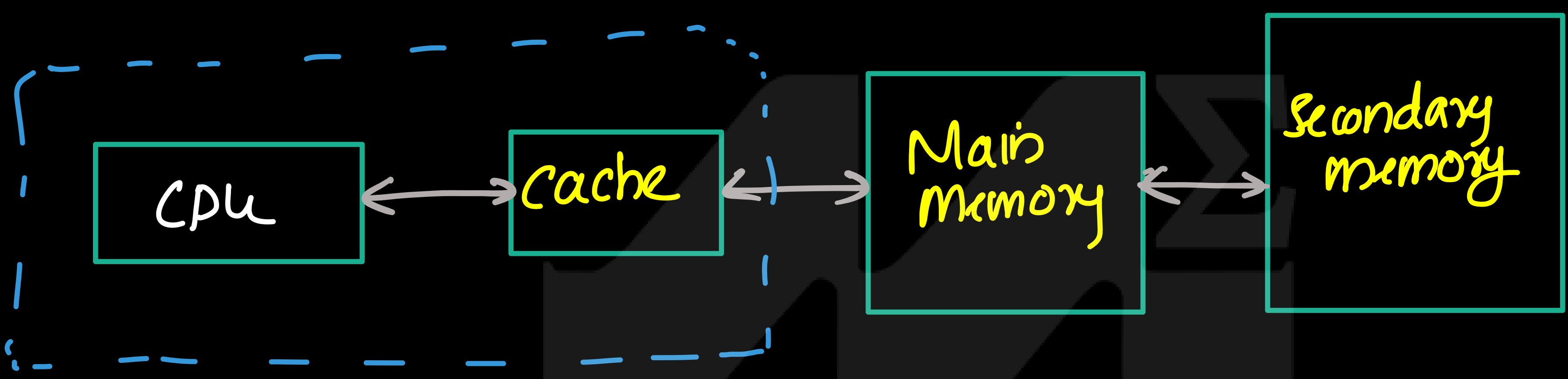
Larger memory is cheaper

Memory hierarchy

- Memory hierarchy is used to bridge gap between the fast cpu and slow large memories. by organizing storage into different levels.
- Most frequently used information is kept in the fastest memory closest to the cpu

memories

1. Cache
2. Main Memory
3. Secondary memory.



processor chip.

→ the data and instructions that are required during the processing of data are brought from secondary memory and stored in the main memory.

→ During the process, instruction data from the main memory should bring to cpu. it will take more time and affect the speed of processing of computer

- So a high speed memory placed b/w cpu and and main memory. Cache memory store more frequently used data and make it available to cpu at a fast rate.

→ During the process, cpu first check cache for the required data, . If data is not found in cache

then it look at the Main memory. and execute the instruction

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→ If required information is not present in the main memory the operating system bring the required information from secondary to main memory. CPU take the data and execute the instruction

— Computer store the most commonly used instructions and data in a faster but smaller memory called cache. Cache is usually built on the same chip as processor. Cache speed is comparable to the processor speed.

— If processor requests data that is available in the cache, it is returned quickly. this is called cache hit. Other wise processor retrieves the data from main memory. this is called cache miss.

Temporal Locality and Spacial Locality.

Note

Temporal locality.

- Temporal locality means that if a memory location (data or instruction) is accessed once. then it is likely to be accessed again within a short period of time.
- So keep the content of recently accessed memory location (data / Instruction) in higher level of memory (cache).

Spacial locality

- spacial locality means that, if a memory location is accessed, then near by memory locations are likely to be accessed soon

→ when a data accessed from the memory location, bring data from near by memory location into higher level of memory hierarch.

Suppose a computer system has a memory organization with only two levels of hierarchy, a cache and main memory. What is the average memory access time given the access times and miss rates in Table

Table 8.1 Access times and miss rates

Memory Level	Access Time (Cycles)	Miss Rate
Cache	1	10%
Main Memory	100	0%

$\frac{10}{100}$ $\underline{0.1}$

$\frac{0}{100} = \underline{0}$

CPU CM MM

H_1 H_2

T_1 T_2

AMAT. T, MM

$$AMAT = \underline{H_1 T_1} + \underline{(1-H_1) H_2} \, \underline{T_1 + T_2}$$

$T_1 = 1$

$T_2 = 100$

$H_1 = \underline{\underline{0.9}}$

$H_2 = 1$

$= 0.9 \times 1 + (1-0.9) \cdot 1 \, (100+1)$

$= 0.9 + 0.1 \times 101$

$= 0.9 + 10.1$

$= \underline{\underline{11}}$ cycle

$HR = 1 - MR$

$H_1 = 1 - M_1$

$= 1 - 0.1$

$= 0.9$

$H_2 = 1 - M_2$

$= 1 - 0 = \underline{\underline{1}}$

Memory Performance (Note)

- The performance of a memory system is measured using the following parameter
  ① Hit Rate ② Miss Rate ③ average memory access time

- Hit and Miss

processor ⟷ cache ⟷ main memory ⟷ secondary memory

- when processor requests instructions/data, it first check the memory closest to cpu (cache). If requested information is present in the memory (cache). then it is called cache hit. In this case cpu gets the data quickly.

- If requested information is not found in the memory (cache). then it is called miss. (cache Miss). then data must be fetched from low level memory (main memory)

If Requested information present in the memory, then it is called Hit. then processor get the information quickly.

If Requested information is not present in the memory. then it is called Miss. Then fetch the information from the low level memory

Hit Rate (H)

$$\text{Hit Rate} = \frac{\text{number of hits}}{\text{Total number of memory access}}$$

$$\text{Hit Rate} = 1 - \text{Miss Rate}$$

Miss Rate

$$\text{Miss Rate} = \frac{\text{number of misses}}{\text{Total number of memory access}}$$

$$\text{Miss Rate} = 1 - \text{Hit Rate}$$

Suppose a program has 2,000 data access instructions (loads or stores) and 1,250 of these requested data values are found in the cache. The other 750 data values are supplied to the processor by main memory . What are the miss and hit rates for the cache?

Total No: of memory access = 2000

No: of Hits (Cache) = 1250

no : of Misses (Cache) = 750

$$\text{Hit Rate} = \frac{\text{No: of Hit}}{\text{Total no: of mem access}} = \frac{1250}{2000} = 0.625 = 62.5\%$$

$$\text{Miss Rate} = 1 - HR = 1 - \frac{1250}{2000} = 37.5\%$$

③ Average Memory access time

Consider a 3 level memory system (M_1, M_2, M_3) with hit ratio H_1, H_2, H_3 respectively. access time for each memory is t_1, t_2, t_3

	Cache	main Memory	Secondary memory.
Hit Ratio	H_1	H_2	$H_3 = 1$
Time	t_1	t_2	t_3.

→ CPU request data, If it is not present (miss) in the Cache , then check in the main memory. If data is not not present in the main memory , then check in the secondary Memory. data will be available in the

Secondary memory so hit rate ($H_3 = 1$).

→ Average memory access time (AMAT)

$$\begin{matrix} H_1 & H_2 & H_3 = 1 \\ t_1 & t_2 & t_3 \end{matrix}$$

$$AMAT = \underset{H}{(\ )}\underset{t}{(\ )} + \underset{H}{(\ )}\underset{t}{(\ )} + \underset{H=1}{(\ )}\underset{t}{(\ )}$$

$$= H_1 t_1 + (1-H_1) H_2 [t_1 + t_2] + (1-H_1)(1-H_2) H_3 (t_1 + t_2 + t_3)$$

$$\boxed{AMAT = H_1 t_1 + (1-H_1) H_2 (t_1 + t_2) + (1-H_1)(1-H_2)(t_1 + t_2 + t_3)}$$

Average memory access time is the average time the CPU spend to get data/instruction from the memory system.

Suppose a computer system has a memory organization with only two levels of hierarchy, a cache and main memory. What is the average memory access time given the access times and miss rates in Table

Table 8.1 Access times and miss rates

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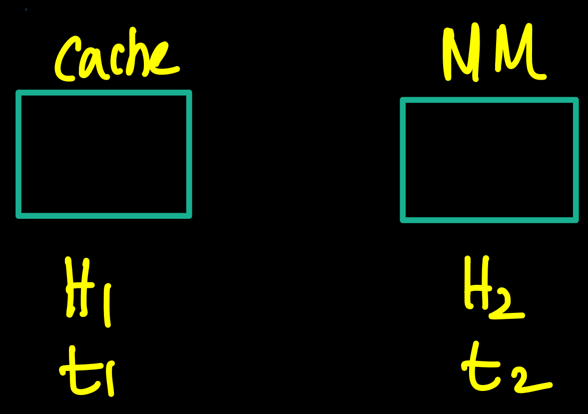


$H_1 = 1 - \text{miss rate} = 1 - 0.1 = \underline{\underline{0.9}}$

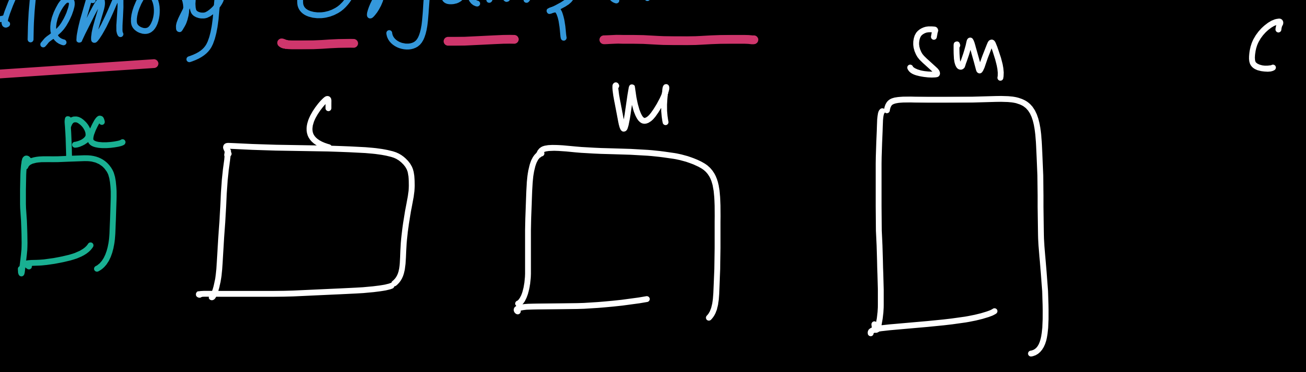
$H_2 = 1 - \text{miss rate} = 1 - 0 = \underline{\underline{1}}$

$t_1 = 1$ (cycle)

$t_2 = 100$ (cycle)

$$AMAT = H_1 t_1 + (1-H_1)(H_2)(t_1 + t_2)$$
$$= 0.9 \times 1 + 0.1 \times 1 \times 101 = 0.9 + 10.1 = 11 \text{ cycles.}$$

Main Memory Organization



$K \rightarrow 2^{10}$
$M \rightarrow 2^{20}$
$G \rightarrow 2^{30}$

$2^5 = 32$
$2^6 = 64$
$2^7 = 128$
$2^8 = 256$
$2^9 = 512$
$K \leftarrow 2^{10} = 1024$
$M \leftarrow 2^{20}$
$G \leftarrow 2^{30}$

Msize = 4 GB → 1By

$= 2^2 \times 2^{30}$ Byte

$= 2^{32}$ Byte $= 4.29 \times 10^9$

4MB → $2^{20} \times 2^2$

Main Memory

Block: W0, W1, W2, W3, W4, W5, W6, ..., W63

Block → $\boxed{W_0, W_1, W_2, W_3}$
$\boxed{W_4, W_5, W_6, W_7}$

Main Memory Size = MMS = 64B

MS = 64B

1Byte = 1 word.

MS = 64 × 1Byte
= 64 word

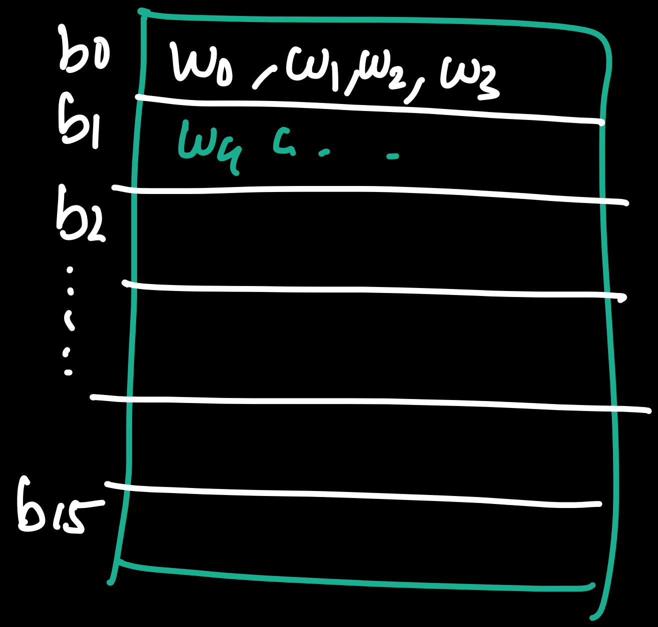
Block size = 4 words   0 — 5
↳ no: of words in a Block

64 word: 4W: W_0, W_1, W_2, W_3 → block [0]

Total no. of Block $= \frac{64W}{4W}$
$= 16$

$= \frac{MS}{BS}$

64 wor



$BS = 4B = 4$ word

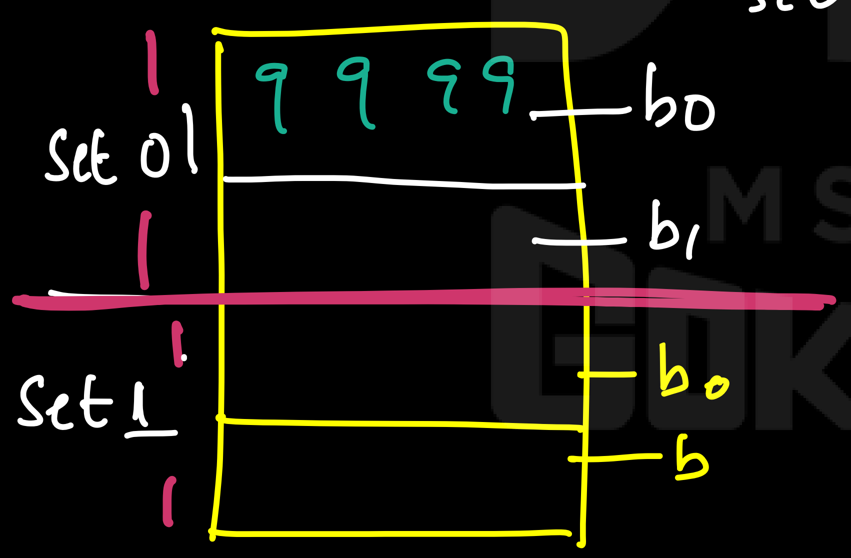
Block No	Main Memory (64 bytes)			
0	W 0	W 1	W 2	W 3
1	W 4	W 5	W 6	W 7
2	W 8	W 9	W 10	W 11
3	W 12	W 13	W 14	W 15
4	W 16	W 17	W 18	W 19
5	W 20	W 21	W 22	W 23
6	W 24	W 25	W 26	W 27
7	W 28	W 29	W 30	W 31
8	W 32	W 33	W 34	W 35
9	W 36	W 37	W 38	W 39
10	W 40	W 41	W 42	W 43
11	W 44	W 45	W 46	W 47
12	W 48	W 49	W 50	W 51
13	W 52	W 53	W 54	W 55
14	W 56	W 57	W 58	W 59
15	W 60	W 61	W 62	W 63

Main Memory
W 0
W 1
W 2
W 3
W 4
W 5
W 6
W 7
W 8
W 9
W 10
W 11
W 12
.
.
.
.
.
.
.
.
.
W 55
W 56
W 57
W 58
W 59
W 60
W 61
W 62
W 63

Cache Memory organi

CS = 16 Byte (16 words)

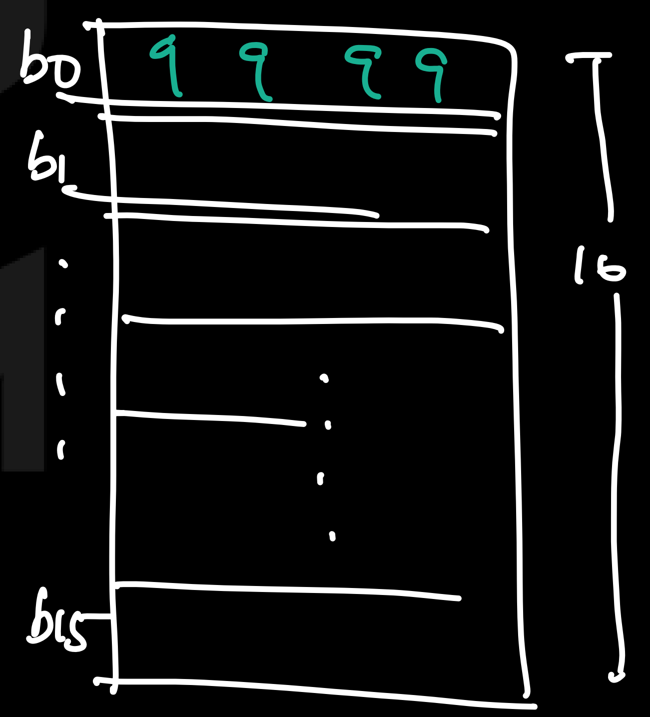
set



Cache is organised into sets

MM

64 Byte (64 words)



Block size = 4W

Block size of cache = Block size of MM

BS = 4 word / 4 Byte

Total No of set = 2

Total no. of Block $= \dfrac{\text{total MS}}{\text{BS}} = \dfrac{16 \text{ word}}{4W} = 4$

Total No. of set $= \dfrac{\text{total no. of B}}{\text{no. of Block in a set}} = \dfrac{4}{2} = 2$

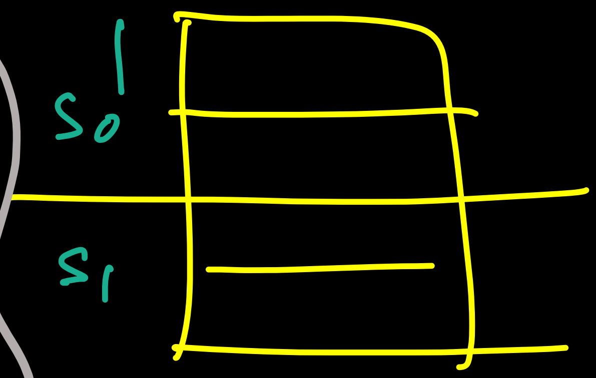
$\dfrac{4}{2} = 2$

Cach

MS = 16B /16W

BS = 4B/ 4W

no: of set = 2



MM

MS = 64B /64W

BS = 4B /4W

no: of Block = $\frac{64}{4} = \underline{\underline{16}}$